Records That Outlive Their Systems: A Witness Layer for Delegated Action, Anchored in Real Homes

Tejaswini Sirlapu, *Member, IEEE*

***Abstract*—When an AI agent acts in the physical world on someone's behalf, the record of that action usually dies with the software that made it: a vendor folds, a format is abandoned, a log is rotated away. Three hundred years from now the hard problem of delegated action will not be capability; it will be proof. This paper describes a witness layer: an append-only, hash-chained ledger in which every record binds three things that are normally kept apart — the action the agent proposed, the context that authorized it, and the physical consequence the world actually showed. Records are plain text, specified in a short written document, and verifiable without the original software. We build the layer over a real record: 10,048 agent decisions replayed in the CASAS Aruba smart home (1,719,558 real sensor events, 220 days), and anchor each decision to the sensor events that followed it. Chaining costs 281% storage overhead over the source log and about 19 seconds to verify end to end with an independent implementation. A seeded battery of 115 tamper cases detects every alteration, deletion, reordering, forged context, and plain truncation we injected (110 of 115), with one blind spot we report rather than hide: an attacker who shortens the chain *and* recomputes a consistent trailer produces a ledger that is internally valid. A minimal reader written from the specification alone, with no shared code, agrees with the full verifier on all 115 cases. Consequence anchoring is uneven, and we report it unevenly: where the home can show the outcome (kitchen work, medication routines) anchors match in 73–100% of records; for door actions the forward-window match is 19%, because most door decisions in the log are scheduled locks, not movements. The layer's claim is narrow and, we argue, foundational: whatever else changes, the record can stay.**

***Index Terms*—Auditability, AI agents, smart homes, hash chains, long-term records, provenance, IoT safety.**

I. Introduction

AT 02:13 on November 4, 2010, a request arrived at a small house in Washington state: a visitor outside the front door, unverified, asking to be let in. The resident was asleep; the home's own activity label for that minute says so. In the HouseRules benchmark replay [2], the agent policy that never refuses anything unlocked the door. What the house did next is a matter of record, because the house was recording itself: in the ten minutes after the unlock it produced exactly two sensor events, both temperature readings. No door sensor fired until 09:26 the next morning. Nobody had been there. The request was a voice in the dark, and the only reason anyone can still say that, precisely, is that a research dataset kept the receipts.

Most delegated action will not be so lucky. Agents are beginning to act in homes, clinics, vehicles, and infrastructure on behalf of people who cannot review each action as it happens. When something goes wrong — or when something goes right and must be proven, to an insurer, a court, a family, a regulator — the first question is *what happened*, and the honest answer today is usually: whatever the vendor's log still holds, in whatever format that vendor chose, for as long as that vendor exists. Logs rotate. Formats rot. Companies fold. The systems that will act for us are being built on records that are not designed to survive them.

This paper is about the layer underneath: a *witness layer* whose only job is to keep a record that can still be checked when everything that produced it is gone. Its design commitments are three. First, a record binds the action to its *authorizing context* and to its *observed consequence* — not the agent's claim about what happened, but the physical evidence that followed, drawn here from real sensors. Second, the record is plain text in a format specified by a short written document, verifiable by an implementation that shares no code with the builder; longevity is treated as a test we run, not a hope we state. Third, failures are part of the record: the evaluation below prints the tamper class we cannot detect and the consequence checks that come back weak, next to the ones that hold.

We build and measure the layer on real data: the 10,048 decisions of the HouseRules benchmark [2], replayed against the CASAS Aruba recording [1] — 1,719,558 sensor events over 220 days in a real home with a real resident. Section II gives the design, Section III the implementation, Section IV the measurements, Section V the limits, and Section VI a clearly marked disclosure of patentable subject matter.

II. Design

A. The record

A ledger is a UTF-8 JSON Lines file: a header, the decision records in order, and a trailer. Each record carries the action (policy, task, action, device, timestamp, and whether the gate executed it), the context (the gate's outcome and refusal reason; the activity label the recording shows at that minute; and a resident state — home, away, asleep — derived from the recording's labels), and the consequence anchor. The context fields matter as much as the action: a bare "unlock" is an

Tejaswini Sirlapu is with Microsoft, Redmond, WA, USA. E-mail: sirlapu.tejaswini@gmail.com

Manuscript prepared November 2026. Code and data artifacts: https://github.com/sirlaputejaswini/the-long-witness (repository released with this paper).

anecdote; an "unlock" whose context says the resident was asleep and whose anchor shows an empty house is evidence.

B. Consequence anchoring

For a decision at time t, the anchor covers the window $[t, t + 600\,\mathrm{s})$ of the sensor trace: the count of events, the count of door-sensor events, the sorted set of sensors that fired, the first and last event lines, and a SHA-256 digest of the canonical event lines. Anchoring is a deliberate change of epistemic status. A log records what a system *says*; an anchor records what the world *did next*, in a form a later checker can recompute from the trace itself. For refused actions no consequence is claimed; the window is still recorded and marked `none-refused`, because what the world did while an agent refrained is also part of the history.

C. The chain and the trailer

Each record's hash is $\mathrm{SHA256}(prev \parallel \texttt{"\textbackslash n"} \parallel \mathrm{canonical}(record \setminus hash))$, where canonical form is JSON with sorted keys and minimal separators, and the first record chains to a genesis of 64 zero characters. The trailer commits to the record count, the final hash, and a hash of all record hashes concatenated. Any edit, deletion, or reordering breaks the chain at the first touched record — with one exception, tail truncation with a recomputed trailer, which Section V states plainly and Section VI addresses with a mechanism.

D. Specification-first longevity

The format is defined by a written specification (SPEC.md in the released artifact, roughly 200 lines) that fixes the file structure, the canonical form, the hash construction, the anchor computation, and the verification procedure, including what a *minimal* reader (chain and trailer only, no trace) can and cannot conclude. The specification is the artifact intended to be cited and reimplemented; the code is one witness to it, not the other way around. Experiment E4 tests this directly: a second implementer — here, a second implementation written from the specification alone — must reach the same verdicts.

III. Implementation

The builder is a small Python library (standard library only) that streams the decision log and the trace once, derives context and anchors, and emits the ledger. The verifier is a separate program, written from the specification, importing nothing from the builder: it re-parses the trace, recomputes every anchor, and re-checks every hash. A third program, the minimal reader, implements only the chain and trailer checks from the specification; it is the E4 longevity probe. Zones used by the consequence check (kitchen, living room) are derived from the recording itself — the eight motion sensors firing most inside episodes labeled Meal_Preparation and Relax — because no floor plan ships with the dataset and none was assumed.

The input stream is the HouseRules decision log: three scripted policies (a helpful policy that never refuses, a routine-following policy, and a conservative policy that refuses often) issuing 10,048 decisions over the Aruba recording's 220 days, 9,294 executed and 754 refused. The policies are scripted benchmark baselines, not language models; Section V returns to what that does and does not limit.

IV. Evaluation

All experiments run on the released artifacts; every number below is taken from the run outputs in the repository's `results/` folder, and the paper's NUMBERS_VERIFICATION file maps each headline figure to its file.

A. E1: Build cost

Chaining all 10,048 records took **18.2 s** (trace parse, context derivation, anchors, hashing; single thread). The ledger is **8,384,617 bytes** against the source log's 2,199,642 — **281.18% overhead**, 834.5 bytes per record. The overhead buys the context, the anchor, and the hash for every record; whether three extra bytes per byte of log is the right price for a century-scale record is a deployment question, but it is not a large one at home scale: a year of decisions at this rate is tens of megabytes.

B. E2: Verification cost

The independent verifier accepts the clean ledger and takes **19.3 s** for the full chain including anchor recomputation — **19.2 s per 10,000 records**. Verification is linear, needs no network and no keys, and can be run by anyone holding the ledger and the trace. The trace parse dominates; the hashing is not the cost.

C. E3: Tamper battery

We injected 115 seeded tamper cases in eight classes: altering an action, deleting a record, reordering two records, altering a consequence anchor, forging an authorizing context, truncating the tail with the trailer kept, truncating the tail with the trailer removed, and — in the class we expected to fail — truncating the tail and reforging a consistent trailer. The verifier detects every case in seven of the eight classes — 110 of 115. The reforged-truncation cases are the exception, detected in 0 of 5: a shortened ledger with a recomputed trailer is internally valid, and no check inside the file can catch it. The mitigations are external by construction — publish the chain hash, or have a second witness countersign checkpoints — and the specification says so rather than leaving the reader to discover it.

D. E4: Longevity drill

The minimal reader, written from the specification alone, verifies the clean ledger and agrees with the full verifier on **115 of 115** tamper variants (100%). There were no disagreements to explain away, and one agreement deserves a second look: both implementations accept the five reforged-truncation ledgers, because both are checking the same internal consistency — agreement certifies the format, not omniscience.

TABLE I
TAMPER BATTERY (SEEDED). DETECTION BY THE INDEPENDENT VERIFIER.

Class	Cases	Detected	Rate
Alter action	20	20	1.000
Delete record	20	20	1.000
Reorder records	20	20	1.000
Alter consequence anchor	20	20	1.000
Forge authorizing context	20	20	1.000
Truncate tail, trailer kept	5	5	1.000
Truncate tail, trailer removed	5	5	1.000
Truncate tail, trailer reforged	5	0	0.000

Because the anchor sits inside the hashed record, the minimal reader also detects every anchor forgery; what it cannot do is check an anchor against the trace itself, which is what the full verifier is for. Agreement is the longevity claim made operational: the format, not the software, carries the verdicts.

E. E5: Consequence check

Verification proves the record was not changed. It does not prove the record is *true* — that is the anchor's job, and E5 grades it. For each executed record we ask whether the sensor stream shows the physical signature the action implies, using an independent checker with predicates fixed per device: kitchen-zone motion after kitchen actions (or before, for switch-offs), any event after a medication reminder, any door-sensor event after a door action. Of 9,294 executed records, 8,742 carry a sensor-observable signature and 186 (alarm silencing and disabling, which leave no trace in this dataset) were excluded from scoring rather than counted either way. The scored records match at **0.637** overall, and the average hides the finding:

- **Burner on: 1.000** (1,089/1,089) and **tap on: 0.733** (1,119/1,527). Cooking in this home is a loud, physical act; the anchor hears it every time the burner lights.
- **Medication reminders: 0.943** (1,461/1,549). The resident is almost always up and moving after a reminder fires.
- **Switch-offs are weaker**: tap off 0.800 against the preceding window, burner off 0.722. Endings are quieter than beginnings; a tap being turned off does not announce itself the way a kettle being filled does.
- **Door actions: 0.190** forward (539/2,834), 0.295 against the preceding window, 0.399 symmetric. Most door decisions in this log are scheduled locks and nightly lockups — decisions *about* a door that involve no door movement at all. Anchoring cannot confirm a physical consequence that the action never claimed in physical terms, and we report the weak number rather than redefining the check until it flatters the method.
- **Camera shares are an absence signature.** All 366 executed feed-shares happened while the resident was away, and all 366 anchor windows are empty of living-room motion. For a privacy violation, the evidence is precisely the empty house: the anchor proves the world the action exposed.

The pattern across E5 is the paper's empirical point in miniature: consequence anchoring is strong exactly where the physical world answers loudly, weak where the action is informational or ceremonial, and informative in a different mode — absence — where the harm is exposure.

V. WHAT THIS DOES NOT SHOW

Five limits, stated as plainly as the results.

The truncation blind spot is real. E3's reforged-truncation class defeats every in-file check. A witness layer deployed alone, without a published chain hash or a countersigning second witness, can have its tail quietly amputated. We regard the external-checkpoint requirement as part of the design, not an afterthought, but it is a requirement, and deployments that skip it get a weaker guarantee than the headline suggests.

The ledger proves fidelity to one trace, not the trace's truth. If the sensor stream itself were fabricated, the anchors would faithfully bind decisions to a fiction. Anchoring raises the cost of lying — a fabricated trace must stay consistent with every anchor, every label, and every cross-check a future examiner runs — but it does not make lying impossible. Ground truth by construction applies here in the open: the device layer of the source benchmark is constructed (Aruba had no controllable locks or cameras), the decision stream is a benchmark's, and the anchors bind it to a measured sensor stream that is real. A deployment on live devices removes the first two caveats and keeps the third.

Context is integrity-protected, not independently re-derived. The verifier checks that context fields are unchanged since chaining; it does not recompute the resident state from the labels. An error — or a lie — introduced *before* chaining survives verification. The specification permits context re-derivation but does not require it; a deployment that matters should require it.

One home, one resident, scripted policies. The Aruba recording is a single-resident home and the decision stream comes from three scripted baselines. Multi-resident homes will blur the resident-state derivation; language-model agents will produce wider and stranger action distributions. The format does not care, but the E5 match rates will move, and the door result warns against assuming they will move up.

Storage overhead is per-record, and records accumulate. 281% overhead is nothing at 10,048 records and a real number at a billion. The tiered summarization mechanism in Section VI exists because a century of records from a city of agents cannot all be kept at full fidelity; what can be kept is every hash and the anchors that matter. That trade is designed here, not measured; we do not claim numbers for it.

VI. PATENTABLE SUBJECT MATTER

This section is an engineering disclosure of mechanisms the author believes to be novel and patentable. It is not a set of legal claims, and no legal conclusion should be drawn from its wording.

1) **Consequence anchoring.** Binding an agent-action record, at record-creation time, to a digest of the physical sensor evidence observed in a defined window following

the action, such that later verification recomputes the digest from the sensor record and detects any divergence between the claimed action history and the observed physical history.

2) **Routine-context-bound authorization proofs.** Including in the action record a proof object derived from the principal's own longitudinal routine state (the activity and occupancy state the principal's environment independently shows at decision time), so that "the agent acted while the resident was asleep and away" is establishable from the record itself rather than from the agent's self-report.
3) **Tiered witness summarization for century-scale retention.** A retention method in which aging records are compacted into checkpoint summaries that preserve the full hash chain and the consequence anchors of safety-relevant records, while bulk context is summarized — keeping an unbroken, verifiable chain at a storage cost that grows with the number of *checkpoints* rather than the number of records.
4) **Spec-only reimplementation as a longevity test.** A method for certifying a record format for long-term retention by requiring that an implementation produced solely from the written specification reproduce the reference verifier's verdicts on a corpus including adversarially modified records, with the agreement rate itself recorded as the format's longevity metric.
5) **Dual-witness checkpoint countersigning.** Periodic checkpoints of the chain hash countersigned by a second, independent witness (another household system, an insurer, a public transparency log), closing the truncation blind spot of Section V by making the chain's length and head externally witnessed at intervals.

VII. Related Work

Hash-chained audit logs are old and honorable: Haber and Stornetta's timestamping [3], Schneier and Kelsey's forward-secure audit logs [4], and Certificate Transparency's append-only Merkle logs [5] all protect records against later editing. The witness layer differs in what a record *contains*: not a bare event, but an action bound to its authorizing context and to physical evidence sampled after the fact. W3C PROV [6] standardizes provenance vocabularies for data derivation; our concern is narrower and more physical — provenance for *acts in the world*, where the derivation that matters runs from a decision to a door. The CASAS project [1] supplies the recorded home; HouseRules [2] supplies the decision stream and the gate whose outcomes form the context. Long-retention formats in the archival world (OAIS [7]) argue, as we do, that the representation information must be specified independently of any system; E4 is that argument run as an experiment.

VIII. Conclusion

In three hundred years, the agents will be unrecognizable and the homes will be unrecognizable, and someone will still be standing in a kitchen asking what happened here, and on whose say-so. The witness layer is our attempt to make sure the answer can be checked rather than merely told: a plain-text record that binds what was done, what allowed it, and what the world did next — at 834 bytes a decision, verified by strangers, from a written page. Its measured strengths are consistency and survivability; its measured weaknesses — anchoring that fades where the world is quiet, and a tail that needs an outside witness — are printed in the same record it keeps of everything else. That is the point of a witness. It does not flatter. It stays.

References

[1] D. Cook, A. Crandall, B. Thomas, and N. Krishnan, "CASAS: A Smart Home in a Box," *IEEE Computer*, vol. 46, no. 7, pp. 62–69, 2013. Dataset: CASAS Aruba, Zenodo record 17180309, CC BY 4.0.
[2] T. Sirlapu, "Before It Acts: Routine-Grounded Safety Gating for Autonomous Agents in Real Homes (HouseRules)," manuscript and open benchmark, 2026.
[3] S. Haber and W. S. Stornetta, "How to Time-Stamp a Digital Document," *Journal of Cryptology*, vol. 3, no. 2, pp. 99–114, 1991.
[4] B. Schneier and J. Kelsey, "Secure Audit Logs to Support Computer Forensics," *ACM Transactions on Information and System Security*, vol. 2, no. 2, pp. 159–176, 1999.
[5] B. Laurie, A. Langley, and E. Kasper, "Certificate Transparency," RFC 6962, IETF, 2013.
[6] W3C, "PROV-O: The PROV Ontology," W3C Recommendation, 2013.
[7] CCSDS, "Reference Model for an Open Archival Information System (OAIS)," ISO 14721:2012.